

Theme 4 - Application of

Note Title

2012/04/19

differentiation

Unit 4.1 Maximum & Minimum
Values

Stewart : Section 4.1 p274 -
p282

Definition 1

Let c be a number in the domain D of a function f .

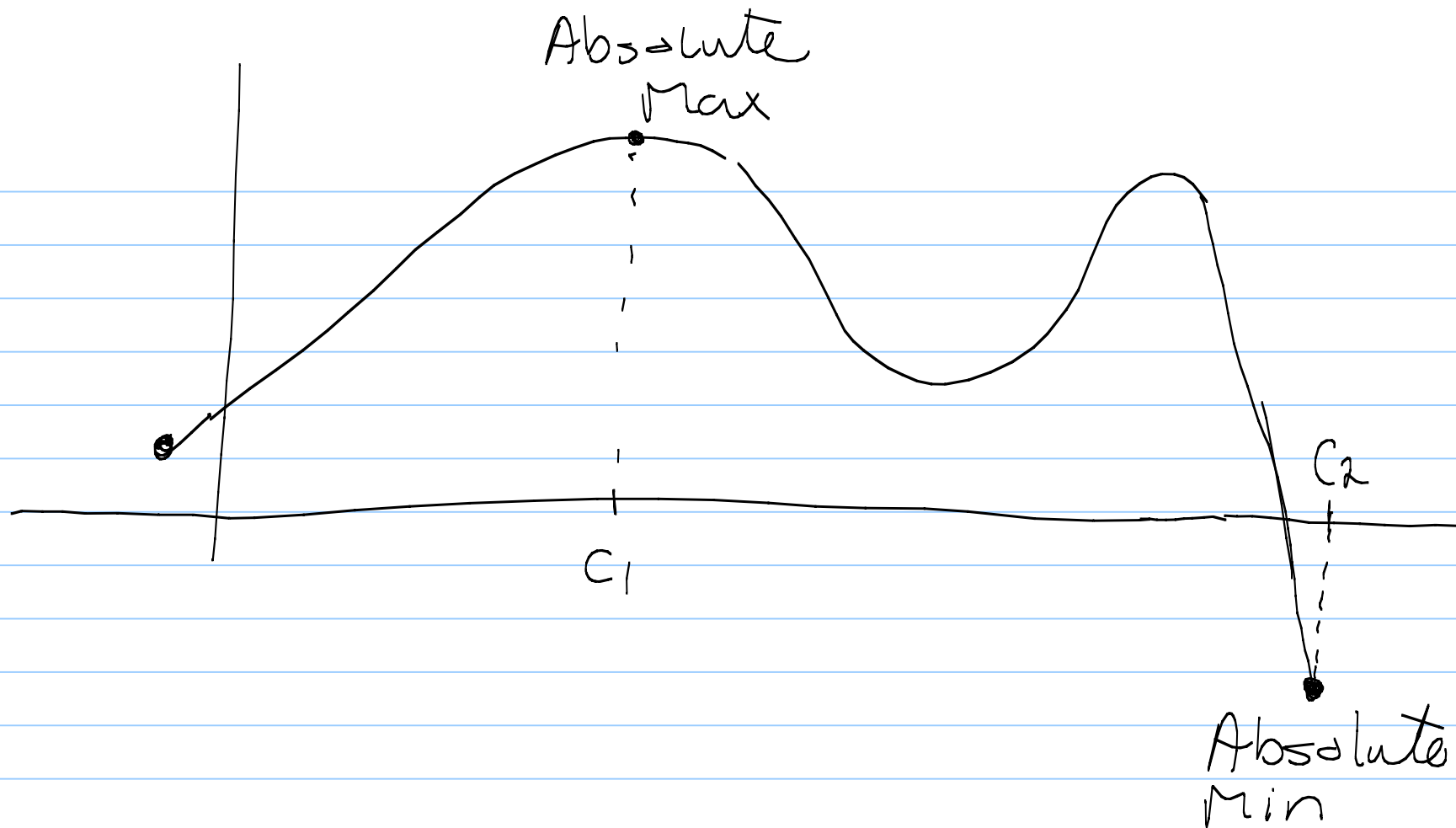
Then $f(c)$ is the

- absolute maximum value of f on D if $f(c) \geq f(x)$ for all x in D

- absolute minimum value of f on D if $f(c) \leq f(x) \forall x \in D$.

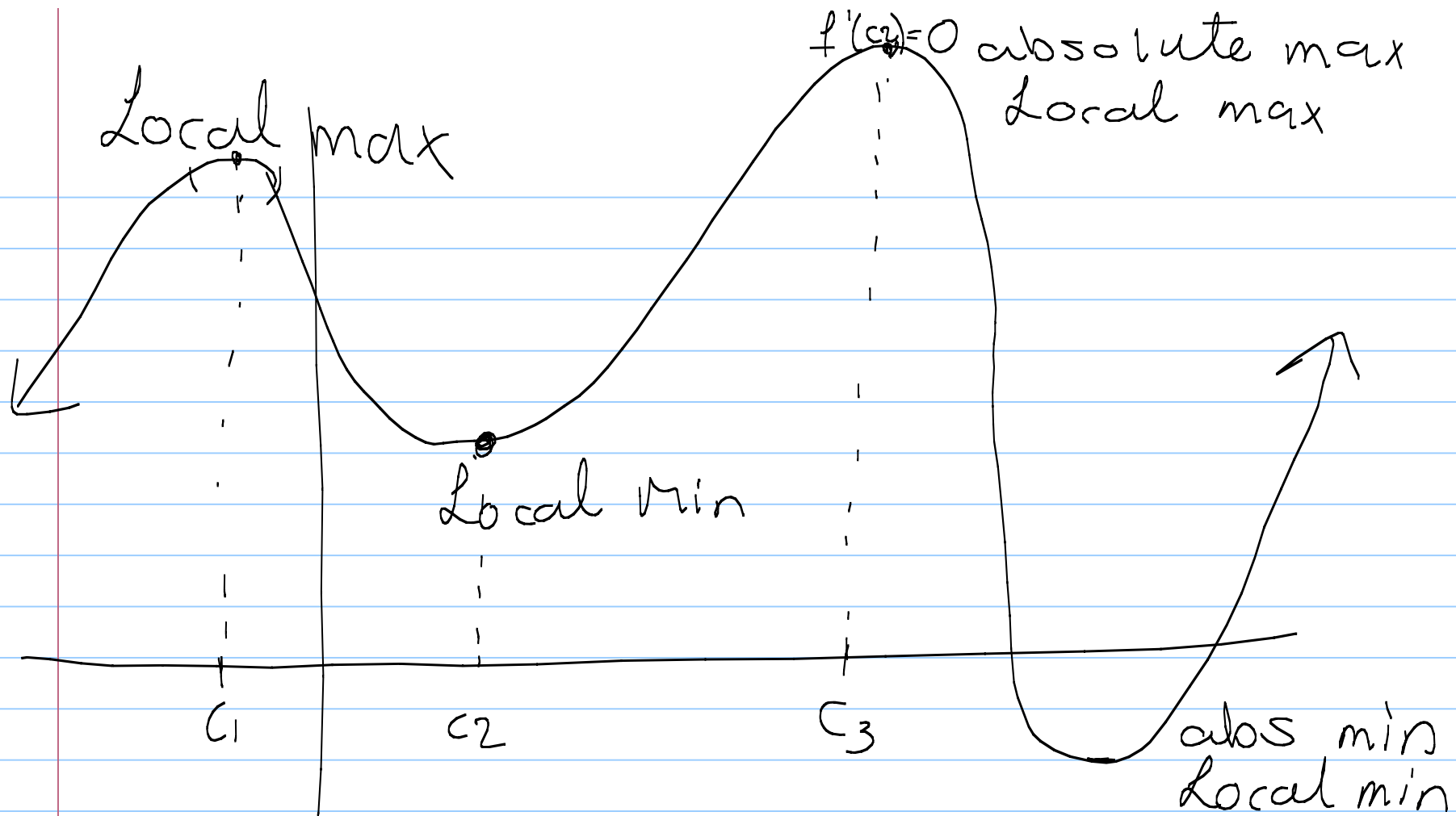
Note

- ⑧ An absolute max/min is sometimes called the global max/min.
- ⑧ The max/min values of f are called the extreme values of f .



Definition 2. The number $f(c)$ is a

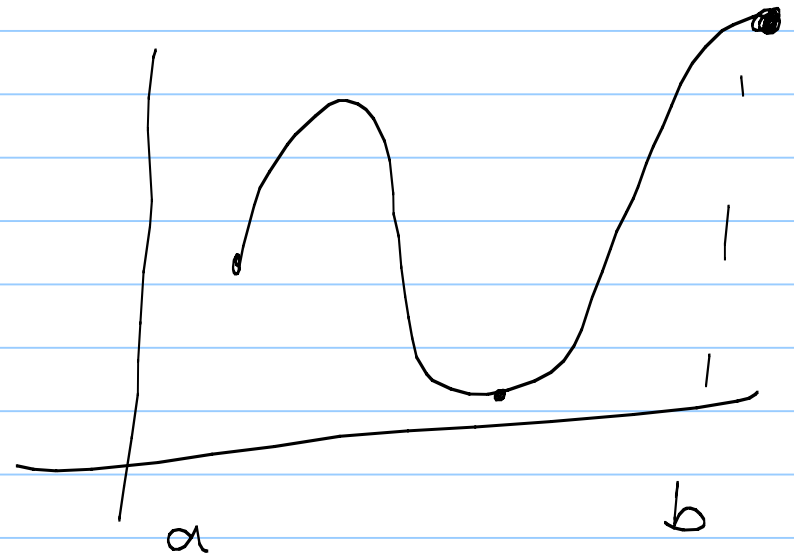
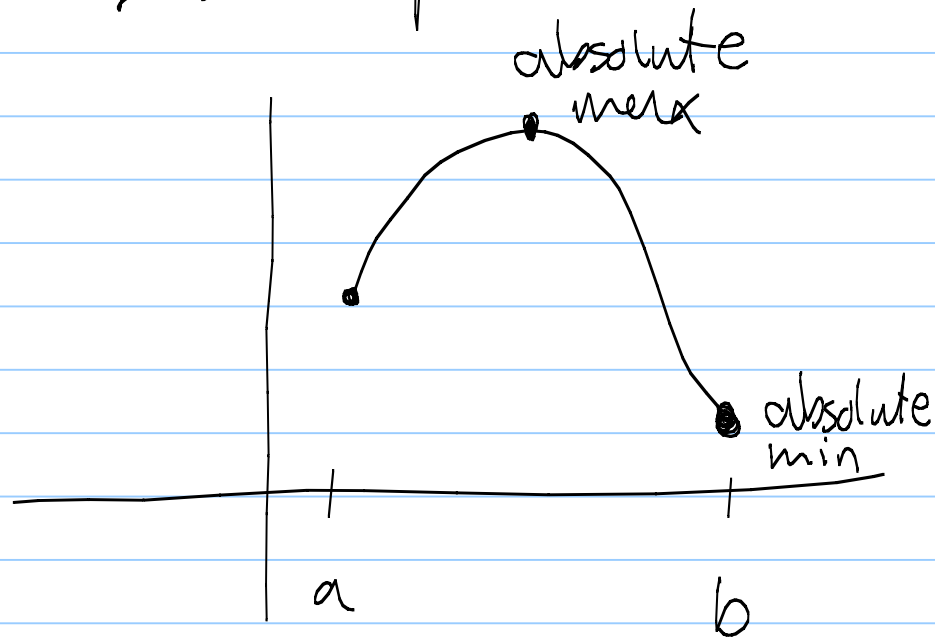
- local maximum value of f
if $f(c) \geq f(x)$ when x is near c . (i.e. in a small interval containing c)
- local minimum value of f
if $f(c) \leq f(x)$ when x is near c .



The Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute min value $f(d)$ at some numbers c and d in $[a, b]$.

Example



∴ When do you know that a function has absolute extremes?

It is continuous on a closed interval.

Fermat's Theorem

If f has a local max or min at c , and if $f'(c)$ exists, then $f'(c) = 0$

Definition A critical number
of a function f is a number
 c in the domain of f such
that either $f'(c) = 0$ or
 $f'(c)$ does not exist.

* A critical number is in the domain of the function.

* A critical number is not a function value

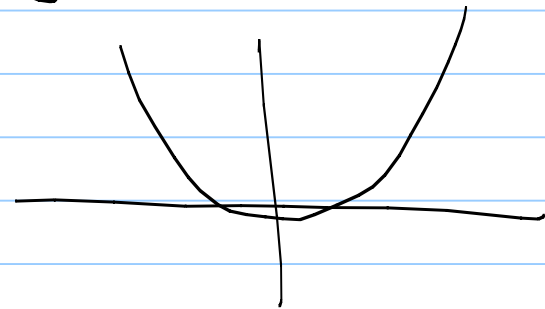
Example 1. Find critical

numbers and decide whether there is a local or absolute max/min by using a sketch.

① $f(x) = x^2$

$$f'(x) = 2x$$

f has a absolute min of 0



at $x = 0$

OR

f has an absolute min

$$f(0) = 0$$

You have to give the
function value and the x -value

Example 2

Find the extremes of

$$f(x) = -x^2 + 1$$

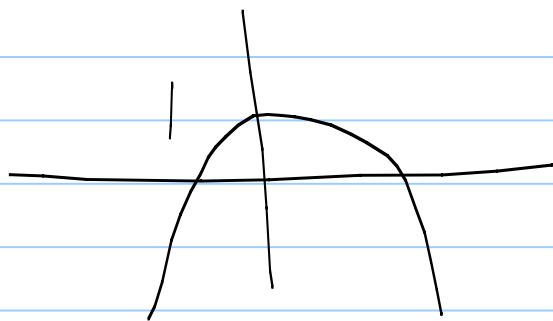
$$f'(x) = -2x$$

$$-2x = 0$$

Sol: f has an absolute
maximum $f(0) = 1$

$$x = 0$$

critical
number



Example 3 Find critical numbers of the function

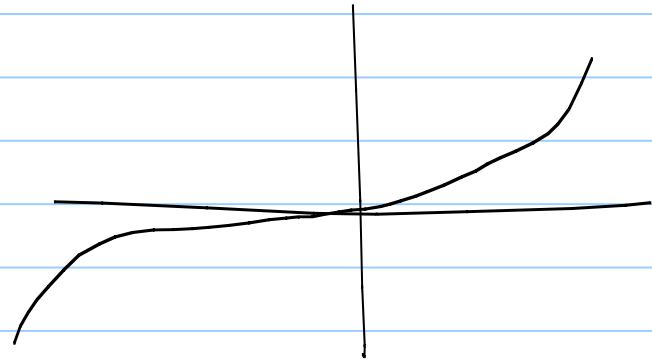
$$f(x) = x^3$$

$$f'(x) = 3x^2$$

$$\Rightarrow 3x^2 = 0$$

$$\Rightarrow x = 0$$

critical number at $x=0$
and no local or absolute extreme

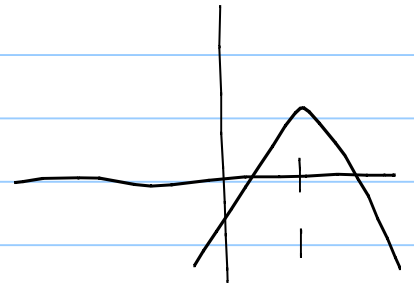


Example 4

$$f(x) = -|x-1| = \begin{cases} -(x-1) & \text{if } x-1 \geq 0 \\ (x-1) & \text{if } x-1 < 0 \end{cases}$$

absolute and local max

at $x=1$ and no local
or absolute min.



Definition . If f has a local max/min at c , then c is a critical number of f .

Homework Find critical values of

$$f(t) = t - \frac{4}{t^2}$$

Theorem If f is continuous on a closed interval, then f has an absolute maximum and an absolute minimum on the closed interval.

The Closed interval method

To find the absolute max and min values of a continuous function f on a closed interval $[a, b]$:

- 1] Find the values of f at the critical numbers of f in (a, b)

2] Find The values of f at the endpoints of the interval

3] The largest of the values from step 1 and 2 is the absolute maximum value, The smallest of these values is the absolute minimum value

Example 1

Find absolute extremes of The function

$$f(x) = x^4 - 2x^2 \text{ on the interval } [-0.5, 2]$$

Critical numbers are 0, 1 and -1 but -1 is not in the interval.

x	$-1/2$	0	1	2
$f(x)$	-0.437	0	-1	8

Absolute max^{of 8} at $x=2$ and absolute min of -1 at $x=1$

Homework

Consider the function

$$f(x) = \frac{\ln x}{x}$$

- i) Find derivative of the function.
- ii) Find critical numbers of the function.
- iii) Use the closed Interval method to find absolute extremes of f on the interval $[1, 5]$

Homework. Consider the function

$$f(x) = x^2 e^{-x}, \quad -1 \leq x \leq 3$$

- i) Find derivative of The function.
- ii) Find critical numbers of The function.
- iii) Use closed interval method to find absolute extremes on $[-1, 3]$.

Homework Find absolute extremes

of

$$f(x) = \sqrt{9 - x^2} \quad \text{on}$$

interval $[-1, 2]$